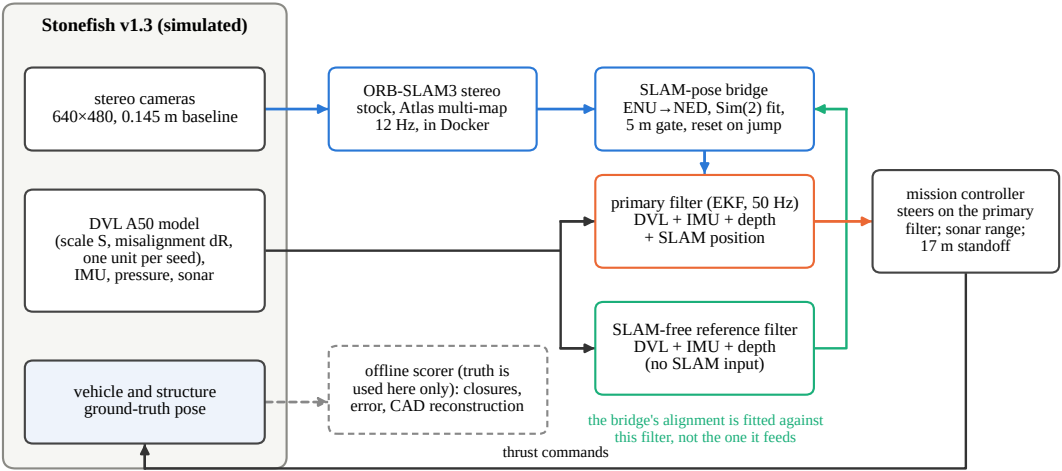
Stonefish v1.3 (simulated)
stereo cameras
640×480, 0.145 m baseline
DVL A50 model
(scale S, misalignment dR,
one unit per seed),
IMU, pressure, sonar
vehicle and structure
ground-truth pose
ORB-SLAM3 stereo
stock, Atlas multi-map
12 Hz, in Docker
offline scorer (truth is
used here only): closures,
error, CAD reconstruction
SLAM-pose bridge
ENU→NED, Sim(2) fit,
5 m gate, reset on jump
primary filter (EKF, 50 Hz)
DVL + IMU + depth
+ SLAM position
SLAM-free reference filter
DVL + IMU + depth
(no SLAM input)
mission controller
steers on the primary
filter; sonar range;
17 m standoff
the bridge's alignment is fitted against
this filter, not the one it feeds
thrust commands